

EXPERIMENTAL STUDY ON HIGH-PRESSURE DROPLET EVAPORATION USING MICROGRAVITY CONDITIONS

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Evaporation of an individual fuel droplet at high pressures and high temperatures has been studied experimentally under microgravity conditions. A suspended *n*-heptane droplet was used in the experiments at pressures in the range of 0.1–5.0 MPa and temperatures varying from 400 to 800 K. Temporal variations of the droplet diameter were measured with a computer-aided image analysis system. Microgravity conditions, which were produced by using 5-m and 110-m drop towers and parabolic flights, were employed to prevent natural convection that complicates the phenomena.

It was observed that dense fuel vapor surrounded a droplet and the droplet surface became obscure at high pressures and high temperatures. The slope of the temporal variations of the squared droplet diameter initially increases but later becomes approximately constant at ambient pressures below the critical pressure of the fuel. At a pressure of 5.0 MPa and temperatures below the critical temperature, the slope becomes less in the latter half of the evaporation lifetime. The ratio of the initial heat-up time to the evaporation lifetime was used as a measure of unsteadiness of droplet evaporation. The ratio is almost independent of ambient temperature at an ambient pressure of 0.1 MPa, but, as ambient pressure is increased, its tendency to rise with ambient temperature becomes noticeable. Corrected evaporation lifetime t_e decreases monotonically as ambient temperature is increased. The slope of its curve becomes steeper as ambient pressure increases. Dependence of t_e on ambient pressure changes according to ambient temperature. Above 550 K, t_e decreases as ambient pressure is increased. Below 450 K, t_e tends to increase as ambient pressure is increased. It is suggested that there exists a certain ambient temperature at which ambient pressure has little effect on t_e .

Introduction

Spray combustion is applied to combustors, for example, liquid rocket engines, diesel engines, jet engines, and industrial furnaces. Operating pressures in such combustors tend to increase and, for some engines, exceed the critical pressures of their fuel. However, there is little useful, basic data on high-pressure spray combustion for designing combustors. Droplet evaporation lifetime is an important parameter for design of combustors because the

evaporation lifetime of the largest droplet in a spray determines the residence time required for complete droplet evaporation in the combustion chamber. Some experimental studies have been performed on droplet evaporation [1–10]. Most of them are for low-pressure, low-temperature conditions and/or forced flow conditions. There are very few experimental studies on high-pressure droplet evaporation except for Matlotsz et al. [8], Hiroyasu et al. [9], and Kadota and Hiroyasu [10]. On the other hand, many theoretical and numerical studies have

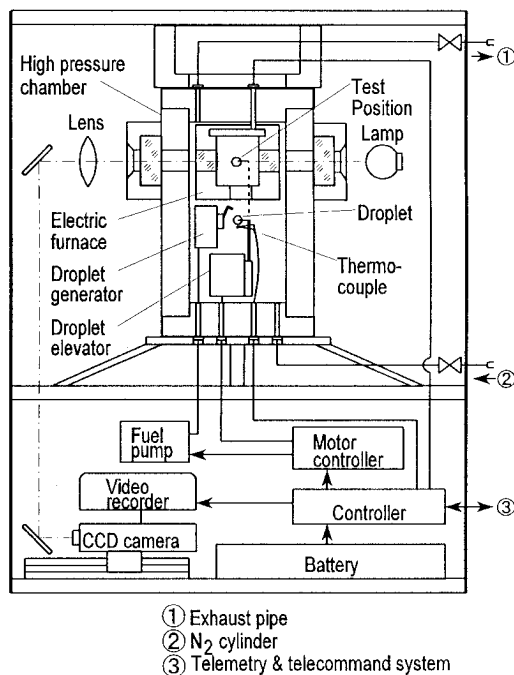


FIG. 1. Schematic of experimental setup.

been performed on high-pressure and high-temperature droplet evaporation [11–17]. To verify the results of these theoretical and numerical studies, experimental data on droplet evaporation is required for high-pressure and high-temperature environments. In most of the theoretical and numerical works, a spherical symmetry model is used. Droplet evaporation under normal gravity conditions is affected by natural convection induced by droplet evaporation itself, and the phenomenon is not spherically symmetric. Its effect becomes dominant as ambient pressure is increased. However, there are no experimental data on spherical droplet evaporation at high pressures and high temperatures. There are some methods to suppress natural convection under high-pressure conditions. One method is to use reduced gravity conditions. Gökalp et al. performed experiments under such conditions and investigated droplet evaporation in forced flows [18]. Their experiments were carried out at low temperatures and atmospheric pressure.

The purpose of this work is to observe the behavior of fuel droplet evaporation that is not affected by natural convection in high-pressure and high-temperature environments. Experiments were performed with an individual suspended droplet under microgravity conditions. Temporal variations of droplet diameter were measured for various ambient pressures and temperatures. Droplet evaporation

lifetimes were obtained from the measured temporal variations of droplet diameter.

Experimental Apparatus

Figure 1 is a schematic of an experimental setup. Evaporation experiments were performed within a high-pressure chamber (inner diameter: 80 mm; inner height: 260 mm). High-pressure gas was fed into the chamber to bring it to the desired test pressure. The chamber was equipped with four windows, each 20 mm in diameter, to allow observation of an evaporating droplet. An electric furnace, droplet generator, and droplet elevator were equipped in the chamber. Since the furnace must be preheated under normal gravity conditions, an electric heater was fitted in the top of the furnace to suppress the natural convection while heating an ambient gas in the electric furnace. A droplet was generated at the tip of a horizontal silica fiber (diameter: 0.15 mm), which was fixed to the movable arm of the droplet elevator. The tip of the fiber was shaped like a ball to prevent the generated droplet from falling down. The initial droplet was at room temperature, and the range of the initial droplet diameter d_0 was 0.6–0.8 mm. At the beginning of observation, the suspended droplet was introduced into the electric furnace and placed at the test position, so that the droplet was rapidly exposed to a high-temperature atmosphere. The required time to introduce a droplet was 0.16 s. Ambient temperature was measured at the position 4 mm distant horizontally from the test position with a chromel-alumel thermocouple of 0.13-mm diameter before the introduction of a test droplet. Nitrogen gas was used as an ambient gas and *n*-heptane (the critical temperature $T_{cr} = 540$ K; the critical pressure $P_{cr} = 2.74$ MPa) was used as a fuel. The ranges of ambient temperature T_a and pressure P_a were about 400–800 K and 0.1–5.0 MPa, respectively.

To realize microgravity conditions, 5-m and 110-m drop towers (ZARM, Bremen University, Germany) and parabolic flights (NOVESPAC, France) were used. Durations of about 1.0, 4.8, and 25 s microgravity conditions were produced, respectively. The free-fall method to produce microgravity conditions with a drop tower is essentially the same as that employed by Kumagai and Isoda for droplet combustion experiments [19,20]. When experiments were performed with the 5-m drop tower, a drag shield was applied to reduce air drag force acting on the experimental setup. When experiments were performed with the 110-m drop tower, the experimental setup was installed into a drop capsule. Inner pressure of the capsule can be kept at atmospheric pressure. After the capsule was set at the top of the drop tube, the free-fall path was evacuated to pressures below 10 Pa. Gravity levels during free fall

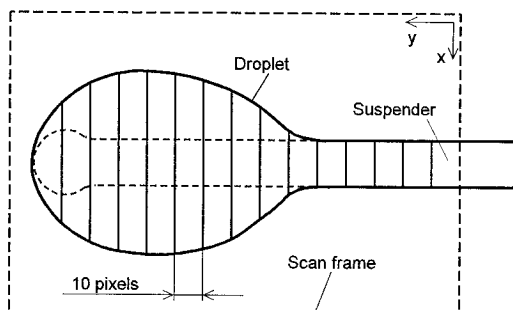


FIG. 2. Schematic of digitized droplet image.

were less than 1/1000 that of the earth's normal gravity. In these two facilities, the droplet generated at the tip of the suspender was introduced into the electric furnace just after the experimental setup started to fall. When experiments were performed by using parabolic flights, the experimental setup was fixed on the floor of an aircraft. The droplet was introduced into the electric furnace about 5 s after the onset of microgravity condition in the aircraft. The maximum g jitter level was about 1/100 that of the earth's normal gravity. The data obtained when a droplet vibrated or fuel vapor flowed were excluded.

Back-lit photography was employed to observe an evaporating droplet. The evaporation process was recorded with a CCD camera and a video recorder. Droplet diameter was measured with a computer-aided image analyzer that was developed for this work. A video picture was converted into a digital picture (512×512 pixels, 256 intensity levels) and divided into 10-pixel lined sections as shown in Fig. 2. Average intensity profiles in the x direction were calculated for each section and the edge of a droplet was determined from them. The position of the edge was defined as the position of the pixel nearest to the suspender among the pixels having the intensity more than $(I_A + 2I_B)/3$, where I_A and I_B are the average intensities of background and droplet, respectively. The volume of a droplet was calculated on the assumption that each 10-pixel lined section has a rotating shape. Droplet diameter was defined as the diameter of a sphere having the same volume of the droplet.

Results and Discussion

Figure 3 shows sequential back-lit photographs of evaporating n -heptane droplets under microgravity conditions. To observe fuel vapor around a droplet, a diaphragm of the CCD camera was removed when these photographs were taken. The first photograph of the sequence shows the droplet just after introduction into the electric furnace. It was observed

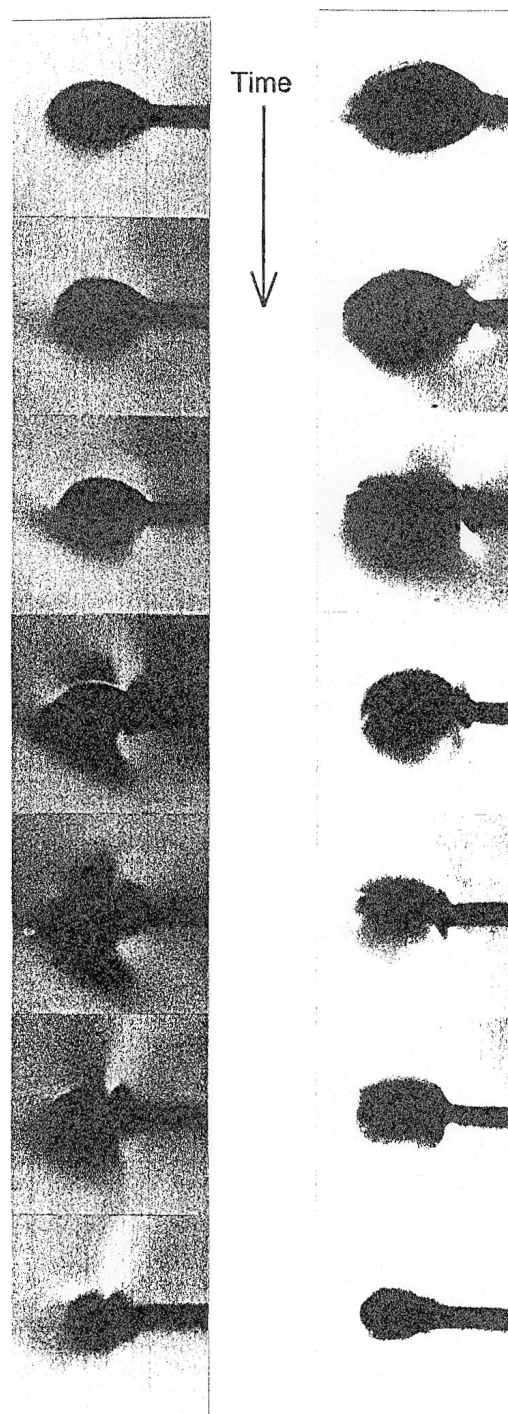


FIG. 3. Evaporating n -heptane droplets under microgravity conditions. (a) $T_a = 622$ K, $P_a = 2.0$ MPa, 10 frames/s; (b) $T_a = 452$ K, $P_a = 5.0$ MPa, 2 frames/s.

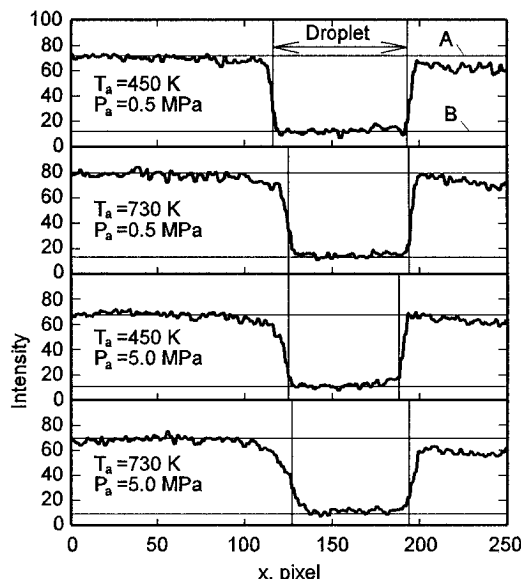


FIG. 4. Intensity profiles of digitized droplet images on the line parallel to the x -axis and passing through the center of a droplet. The images of an n -heptane droplet were recorded at the middle of droplet lifetime under microgravity conditions.

that fuel vapor surrounded a droplet and the outer edge of the droplet became gradually obscured. Under normal gravity conditions, downward flow, which is called natural convection, is induced by buoyancy effect because the n -heptane molecule is heavier than the nitrogen molecule and the gas temperature near the droplet is lower than ambient temperature. Under microgravity conditions, radial vapor flow occurs instead of natural convection, and the vapor layer around a droplet is thicker than under normal gravity conditions. Near and above the critical condition (which means both ambient temperature and pressure are equal to the critical values of the fuel), decrease of the density difference between liquid and vapor in this layer obscures the outer edge of a droplet. Figure 4 shows intensity profiles through the center of a droplet obtained from digitized droplet images. Letters A and B in the figure indicate the average intensities of background and droplet, respectively. It is observed that the slope at the outer edges of a droplet in the bottom graph (supercritical condition) is smoother than the others (subcritical condition). Therefore, measurements of temporal variation of droplet diameter evaporating under supercritical conditions were difficult and measurement results did not have enough repeatability. Results obtained at a pressure of 5.0 MPa and temperatures above 600 K will not be discussed in this paper.

Figure 5 shows variations of squared nondimen-

sional droplet diameter $(d/d_0)^2$ with corrected time t/d_0^2 for various ambient temperatures and pressures. The origin of the horizontal axis is the time when a droplet was placed at the test position. The final droplet diameter was equal to the diameter of the ball at the tip of the suspender (see Fig. 2). The results were not affected by the type of μG facility. At ambient pressures below P_{cr} , $(d/d_0)^2$ decreases slowly in the beginning because of low droplet surface temperature, heat consumption in temperature rise of a droplet, and droplet heat expansion. Gradually, the slope of the curve becomes steeper. The evaporation in this period, which is called the initial heat-up period, is unsteady. After the initial heat-up period, $(d/d_0)^2$ decreases approximately linearly in keeping with d^2 law. The evaporation in this period can be assumed to be quasi-steady. At an ambient pressure of 5.0 MPa ($>P_{cr}$), after a short initial heat-up period, the slope of $(d/d_0)^2$ curve reaches its maximum value and then decreases gradually. This behavior of $(d/d_0)^2$ occurs at lower temperatures under normal gravity conditions [21]. Figure 5f shows the effect of ambient pressure on the $(d/d_0)^2$ curve at ambient temperatures around 655 K. As ambient pressures increase, the ratio of the initial heat-up time to the evaporation lifetime increases and the $(d/d_0)^2$ curve becomes round. This means that droplet surface temperature rises during most or all of the evaporation lifetime in high-temperature and high-pressure atmospheres. For estimation of droplet evaporation lifetime at high temperatures and pressures, it is necessary to take into account unsteadiness of droplet evaporation.

The relationship between $(d/d_0)^2$ and t/d_0^2 is a function of ambient temperature and pressure. Therefore, the ratio of initial heat-up time to evaporation lifetime is also a function of ambient temperature and pressure. This ratio can be a measure of unsteadiness of droplet evaporation. Measurements of the ratio of initial heat-up time to evaporation lifetime Φ_i were made for the conditions when variations of $(d/d_0)^2$ with t/d_0^2 were measured. Figure 6 illustrates the definition of the initial heat-up time and the evaporation lifetime. The solid straight line on the figure is obtained from the plots in the region $(d/d_0)^2 \leq 0.5$ using least squares method. The initial heat-up time t_i is the time at the intersection of this solid line and $(d/d_0)^2 = 1$. The evaporation lifetime t_e is the time at the intersection of the solid line and the abscissa.

Figure 7 shows the results of the measurements of Φ_i as a function of ambient temperature. At an ambient pressure of 0.1 MPa, Φ_i is almost constant within experimental accuracy and the range of its value is about 0.1–0.2. As ambient pressure is increased, Φ_i tends to increase with ambient temperature. At 2.0 MPa, Φ_i is approximately three times as large in the ambient temperature range of 450–750 K. When ambient temperature is higher than

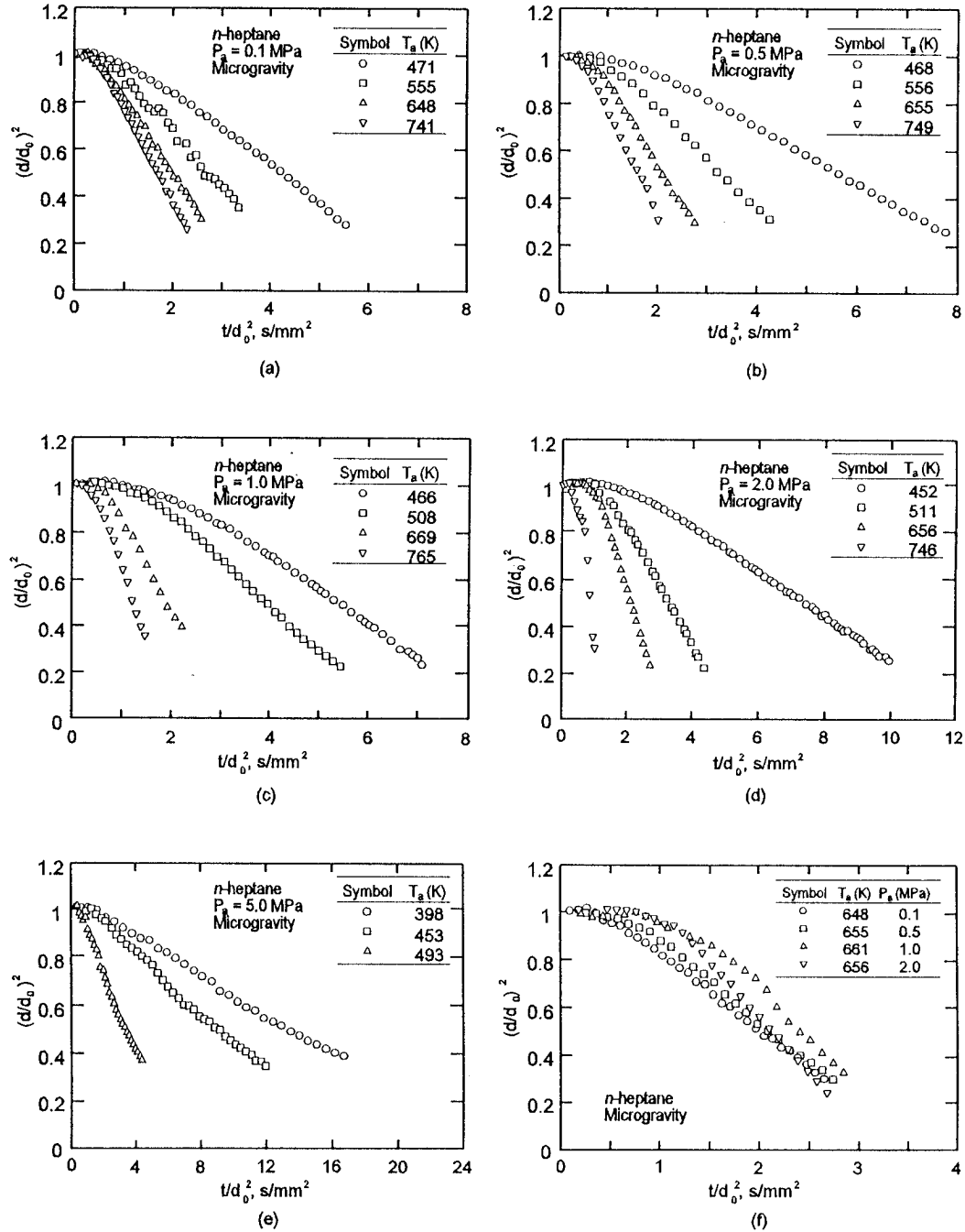


FIG. 5. Variations of squared nondimensional droplet diameter with the time divided by squared initial droplet diameter.

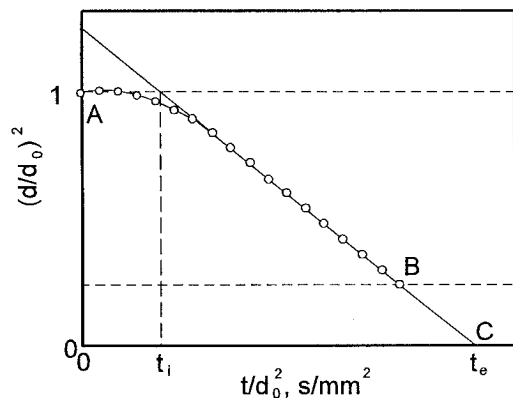


FIG. 6. Definition of initial heat-up time and corrected evaporation lifetime.

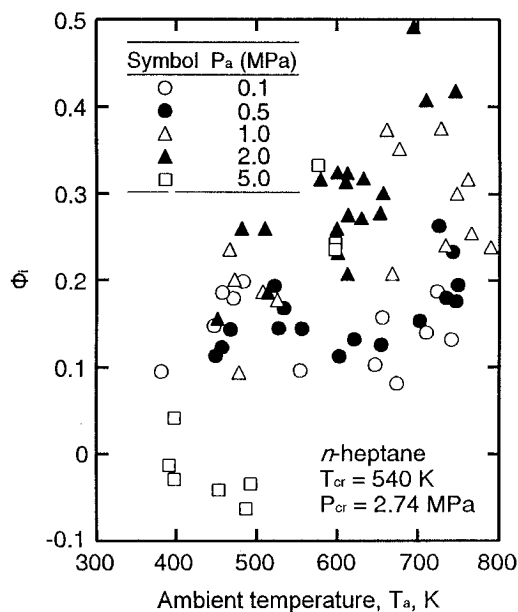


FIG. 7. Effect of ambient temperature on the ratio of initial heat up time to evaporation lifetime.

boiling temperature, wet bulb temperature is saturated to the boiling temperature, and temperature difference between a droplet and an ambient gas increases linearly as ambient temperature increases. It is supposed that the reason Φ_i is almost independent of ambient temperature at 0.1 MPa is that, at temperatures higher than boiling temperature, the time required to heat a droplet up to wet bulb temperature increases proportionally to the required time for complete evaporation as ambient temperature increases. When ambient temperature is less than boiling temperature, as ambient temperature is

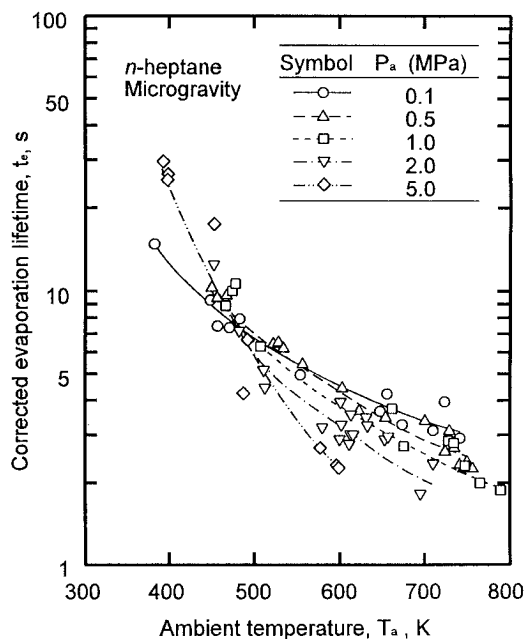


FIG. 8. Variations of corrected evaporation lifetime with ambient temperature.

increased, wet bulb temperature and the temperature difference between a droplet and an ambient gas increase. Increase of wet bulb temperature lengthens time required to heat a droplet and, because of the decrease of latent heat, the time required to make complete evaporation shorter. Consequently, Φ_i increases as ambient temperature increases. The negative values of Φ_i at 5.0 MPa result from $t_i < 0$ and indicate that the slope of $(d/d_0)^2$ curve is reduced in the latter half of the evaporation lifetime.

The evaporation lifetime t_e , as defined previously, is regarded as a characteristic parameter of droplet evaporation and a measure of evaporation rate. Since t_e can be regarded as the lifetime of a droplet 1 mm in initial diameter, a second was used as a unit [22] for t_e , called the corrected evaporation lifetime. Measurements of t_e were made for various ambient temperatures and pressures. Results are shown in Fig. 8. For all ambient pressures, t_e decreases monotonically throughout the ambient temperature range of the present experiments. The slope of the curve becomes steeper as ambient pressure is increased. Dependence of t_e on ambient pressure changes according to ambient temperature. Above 550 K, t_e decreases as ambient pressure is increased. Below 450 K, t_e tends to increase as ambient pressure is increased. This suggests that there is an ambient temperature at which t_e is almost independent of pressure. Its value was estimated at about 480 K for

n-heptane from experimental results. The latent heat of the fuel and the temperature difference between ambient temperature and droplet surface temperature decrease as ambient pressure increases. It is suggested that these two effects balance at about 480 K.

Conclusions

Effects of ambient temperature and pressure on droplet evaporation have been studied experimentally under microgravity conditions. A suspended *n*-heptane droplet was employed for experiments. The temporal variations of droplet diameter were obtained for the ambient pressure range of 0.1–2.0 MPa at ambient temperatures of 400–800 K and for an ambient pressure of 5.0 MPa at ambient temperatures of 400–600 K. Conclusions are as follows:

1. The slope of diameter squared versus time initially increases but later becomes approximately constant at ambient pressures of 0.1–2.0 MPa. At an ambient pressure of 5.0 MPa and temperatures below the critical temperature of the fuel, the slope becomes less in the latter periods of the evaporation lifetime.
2. The ratio of the initial heat-up time to the evaporation lifetime Φ_i was measured as the unsteadiness of droplet evaporation. At an ambient pressure of 0.1 MPa, Φ_i is almost independent of ambient temperature and the range of its value is about 0.1–0.2. As ambient pressure is increased, Φ_i tends to increase with ambient temperature. At 2.0 MPa, Φ_i increases from about 0.15 up to about 0.5 in the ambient temperature range of 450–750 K.
3. Dependence of the corrected evaporation lifetime t_e on ambient temperature was measured for various ambient pressures of 0.1–5.0 MPa. For all pressures, t_e decreases monotonically as ambient temperature is increased. The slope of its curve becomes steeper as ambient pressure increases. Dependence of t_e on ambient pressure changes according to ambient temperature. Above 550 K, t_e decreases as ambient pressure increases. Below 450 K, t_e tends to increase as ambient pressure is increased. It is suggested that t_e is almost independent of ambient pressure at an ambient temperature of about 480 K.

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REFERENCES

1. Katto, Y., JSME Annual Meeting, JSME, 1960.
2. Law, C. K. and Williams, F. A., *Combust. Flame* 19:393–405 (1972).
3. Law, C. K., *Prog. Energy Combust. Sci.* 8:171–201 (1982).
4. Sirignano, W. A., *Prog. Energy Combust. Sci.* 9:291–322 (1983).
5. Kobayasi, K., *Trans. JSME* 20:826–831 (1949).
6. Ranz, W. E. and Marshall, W. R. Jr., *Chem. Engineering Prog.* 48:141–146 (1952).
7. Nishiwaki, N., *Fifth Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, 1954, pp. 148–158.
8. Matlosz, R. L., Leipziger, S., and Torda, T. P., *Int. J. Heat Mass Transf.* 15:831–852 (1972).
9. Hiroyasu, H., Kadota, T., Chida, T., and Imamoto, T., *Trans. JSME* 40:3147 (1974).
10. Kadota, T. and Hiroyasu, H., *Trans. JSME* 44:3884 (1978).
11. Spalding, D. B., *ARS J.* 29:828–835 (1959).
12. Rosner, D. E., *AIAA J.* 5:163–167 (1967).
13. Kotake, S. and Okazaki, T., *Int. J. Heat Mass Transf.* 12:595–609 (1969).
14. Rosner, D. E. and Chang, W. S., *Combust. Sci. Technol.* 7:145–158 (1973).
15. Kadota, T. and Hiroyasu, H., *Trans. JSME* 42:1216–1223 (1976).
16. Curtis, E. W. and Farrell, P. V., *Acta Astronautica* 12:1189–1193 (1988).
17. Curtis, E. W., Hartfield, J. P., and Farrell, P. V., Third International Colloquium on Drops and Bubbles, 1988.
18. Gökalp, I., Chauveau, C., Richard, J. R., Kramer, M., and Leuckel, W., *Twenty-Second Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, 1988, pp. 2027–2035.
19. Kumagai, S. and Isoda, H., *Sixth Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, 1957, pp. 2027–2035.
20. Isoda, H. and Kumagai, S., *Seventh Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, 1959, pp. 523–531.
21. Nomura, H. and Ujiie, Y., *Trans. JSME* 61:1834–1840 (1995).
22. Feath, G. M., Dominicus, D. P., Tulpinsky, J. F., and Olson, D. R., *Twelfth Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, 1969, pp. 9–18.